

Nikolaos Goutzoulis

EDUCATION

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Columbia University

Computer Science MS

• GPA: 3.52/4.00

• Relevant Courses: Unsupervised Learning, Advanced Deep Learning, Algorithms for Data Science

National Technical University of Athens

Electrical and Computer Engineering - Integrated MEng

• Grade: 8.7/10 (top 5% of 400 students)

• Relevant Courses: Deep Learning, Databases, Advanced Algorithms

Aug 2024 - Dec 2025

New York, NY

Jul 2024

Athens, GR

PROFESSIONAL EXPERIENCE

Columbia University

Course Assistant - Analysis of Algorithms

• Guiding 150+ students in mastering dynamic programming, **linear programming**, and **nonlinear optimization** techniques, emphasizing their application in problem-solving and scalable system design.

• Demonstrating leadership and **communication skills** by holding regular Q&A sessions, simplifying complex algorithms, and fostering critical thinking in problem-solving.

Jan 2025 - Present

New York, NY

Columbia University

Graduate Research Assistant – Software Systems Lab

• Collaborating with Meta researchers through **Git** to create a **browser API** in **C++** for Chrome and Firefox by analyzing codebases and focusing on managing budget between first- and third-party sites while preserving user privacy.

• Creating reusable **UniFFI** bindings for **Rust/C++/Python** interoperability, and conducting end-to-end testing, including verifying privacy loss metrics within Google's Attribution Reporting API architecture.

Sep 2024 - Present

New York, NY

EmDoT S.A.

Machine Learning Engineer - Master Thesis Collaboration

• Engineered deep learning models with **PyTorch** using **CUDA**, and machine learning models such as random forests with **Sklearn**, boosting prediction accuracy by 15% for resource allocation KPIs in an insurance company.

• Optimized model performance by 20% and accuracy by performing **feature engineering**, incorporating parameters such as time of day and custom metrics.

• Engaged with cross-functional teams, directly **interacted with customers** and **deployed models** that are currently in industrial use.

Oct 2023 - Jul 2024

Athens, GR

EmDoT S.A.

Data Science Intern

• Designed algorithms for smoothing Bin Fill Level data, using **SQL**, **Tableau**, and statistical models like **Prophet** and **SARIMAX** to predict the trend of the data.

• Reduced noise in data by 20% while maintaining 95% of original signal's variance.

• Collaborated with 2 interdisciplinary teams of 4 engineers, ensuring alignment between software and hardware engineering efforts while meeting client goals.

Jul 2023 - Sep 2023

Athens, GR

PROJECTS AND DISTINCTIONS

Evaluating LLM Resilience to Adversarial Data Poisoning

Columbia University

• Designed a method to assess and quantify the resilience of **RoBERTa** LLM in sentiment analysis against adversarial data poisoning by applying influence function-based perturbations, which revealed a model accuracy drop of up to 15% under structured adversarial attacks.

• Generated text embeddings using BERT and performed **dimensionality reduction** within **Databricks** to facilitate sentiment analysis.

Sep 2024 - Dec 2024

CNN for Variant-Based Gene Expression Prediction

Columbia University

• Developed a parallel computation **CUDA-accelerated** multitasking CNN using **PyTorch** on **AWS** GPU-enabled EC2 instances to predict gene expression from eQTL data across diverse genomes.

Sep 2024-Dec 2024

Aegean Shipping Management Scholarship

Proud of Youth HelleniQ Energy Scholarship

Jul 2024

Jul 2024

SKILLS AND CERTIFICATIONS

Programming Languages: Python, C++, JavaScript, Java, Matlab, SQL, Rust, R

ML Frameworks: Tensorflow, CUDA, PyTorch, Keras, Scikit-learn, Transformers, XGBoost

Cloud, Development and Other Tools: GCP, Databricks, Docker, Kubernetes, PowerBI, API design, Visual Studio,

Matplotlib, Git, Tableau, React, Node.js